Essentials of Probaility Theory

Contents

1 Basics of Probability

Definition 1.1 (Random Experiment)**.** An experiment whose outcome isn't known in advance is called a *Random Experiment.*

Definition 1.2 (Sample Space)**.** The *set* of all possible outcomes of a random experiment is called *sample space.* We will use, S, to denote the sample space.

Definition 1.3 (Events)**.** Any subset, E, of the sample space, S, is called an event.

1.1 Axioms of Probability

1. For any $E \subseteq S$, $0 \leq \Pr(E) \leq 1$
2. $\Pr(S) = 1$
3. For a sequence of pairwise disjoint events, $E_1, E_2, \cdots, E_n$,

$$\Pr\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} \Pr\left(E_i\right)$$

1.2 Simple Propositions

Proposition 1.1. $\Pr(\phi) = 0$

Proposition 1.2. If $E \subseteq F$, then $\Pr(E) \leq \Pr(F)$

Proposition 1.3. $\Pr(E^C) = 1 - \Pr(E)$

Proposition 1.4. For any two events, E and F,

$$\Pr(E \cup F) = \Pr(E) + \Pr(F) - \Pr(E \cap F)$$

1.3 Equally Likely Outcomes

Consider a sample space S, corresponding to some random experiment. If it so happens that we consider all the outcomes of the random experiment to be *equally likely* to occur. Then, for any event E,

$$\Pr(E) = \frac{|E|}{|S|}$$

2 Basic Counting Principles

Proposition 2.1 (**Sum Rule**)**.** If a task t_1 can be done in n_1 ways and another task t_2 can be done in n_2 ways, then the total number of ways to complete *either* task t_1 *or* the task t_2 but *not both* is $n_1 + n_2$.

Proposition 2.2 (**Product Rule**)**.** If a task t_1 can be done in n_1 ways and another task t_2 can be done in n_2 ways, then the number of ways to complete the task t_1 *and* the task t_2 is $n_1 \cdot n_2$.

Definition 2.1. $n! = n \cdot (n-1) \cdot (n-2) \cdots 1$

Remark. $0! = 1$

2.1 Combinations

If we have a set of n people and we want to select k out of them in such a way that the *order* of the selection doesn't matter, then the total number of ways to do that is given by,

$$\binom{n}{k} = \frac{n!}{(n-k)! \cdot k!}$$

$\binom{n}{k}$ is read as "n choose k".

2.2 Permutations

If we have a set of n people and we want to select k out of them in such a way that the *order* of the selection matters, then the total numbers of ways to do that is given by,

$$(n)_k = \frac{n!}{(n-k)!}$$

$(n)_k$ is read as "n falling factorial k"

3 Conditional Probability

Definition 3.1. For any two events E and F, $\Pr(E|F)$, read as, "probability of E given F", is defined as follows,

$$\Pr(E|F) = \frac{\Pr(E \cap F)}{\Pr(F)},$$

such that, $\Pr(F) \neq 0$

Proposition 3.1 (Multiplication Rule of Probability). For any two events, E and F, the multiplication rule of probability states that,

$$\Pr(E \cap F) = \Pr(E) \cdot \Pr(F|E)$$

In a more generalized form, this rule can be stated as follows. For a given sequence of n events, $E_1, E_2, \cdots, E_n$,

$$\Pr\left(\bigcap_{i=1}^{n} E_i\right) = \Pr(E_1) \cdot \Pr(E_2|E_1) \cdot \Pr(E_3|E_1 \cap E_2) \cdots \Pr\left(E_n \middle| \bigcap_{i=1}^{n-1} E_i\right)$$

Definition 3.2 (Partition)**.** For a set S, a sequence of its subsets, $E_1, E_2, \cdots, E_n$, constitute a partition of the set S, if and only if,

- $E_i \cap E_j = \phi$, for $i \neq j$, i.e., the given sequence of events are pairwise disjoint.
- $E_1 \cup E_2 \cup E_3 \cup \cdots \cup E_n = S$

Proposition 3.2 (Law of Total Probability). Consider a partition, $E_1, E_2, \cdots, E_n$ of a set S. Then the law of total probability states that for any event F,

$$\begin{aligned}
\Pr(F) &= \Pr(E_1 \cap F) \cup \Pr(E_2 \cap F) \cup \cdots \cup \Pr(E_n \cap F) \\
&= \sum_{i=1}^{n} \Pr(E_i \cap F) && (E_i \cap E_j = \phi, \text{ for } i \neq j) \\
&= \sum_{i=1}^{n} \Pr(E_i) \cdot \Pr(F|E_i) && \text{(Multiplication Rule of Probability)}
\end{aligned}$$

Theorem 3.1 (**Bayes' Rule**)**.** *For a partition, $E_1, E_2, \cdots, E_n$ of a set S and any event F, the Bayes' Rule states that, for any event, E_j, of the partition,*

$$\begin{aligned}
\Pr(E_j|F) &= \frac{\Pr(E_j \cap F)}{\Pr(F)} && \textit{(Defn. of Conditional Probaility)} \\
&= \frac{\Pr(E_j) \cdot \Pr(F|E_j)}{\Pr(F)} && \textit{(Multiplication Rule of Probability)} \\
&= \frac{\Pr(E_j) \cdot \Pr(F|E_j)}{\sum_{i=1}^{n} \Pr(E_i) \cdot \Pr(F|E_i)} && \textit{(Law of Total Probability)}
\end{aligned}$$

Thus, the Bayes' Rule states that,

$$\Pr(E_j|F) = \frac{\Pr(E_j) \cdot \Pr(F|E_j)}{\sum_{i=1}^{n} \Pr(E_i) \cdot \Pr(F|E_i)}$$

4 Random Variables

Definition 4.1 (Random Variables)**.** A (real-valued) random variable X is a function from the sample space S to the set of all real numbers, $\mathbb{R}$.

A random variable is neither *random* nor a *variable* but it is the way that we use them makes it "reasonable" to call them a "random variable".

Remark. On performing a random experiment if our outcome is s, then we write, $X = x$ to denote that $X(s) = x$.

5 Discrete Random Variables

Definition 5.1 (Discrete Random Variables)**.** A random variable X is said to be a discrete random variable if the *range* of X, denoted as R_X, is a finite set or a countably infinite set.

5.1 Probability Mass Function (PMF) of a Discrete Random Variable

For a discrete random variable X, its PMF denoted as p_X, is a function,

$$p_X : \mathbb{R} \to [0, 1]$$

such that,

$$p_X(x) = \Pr(X = x)$$

Observation 5.1. For a random variable X, let R_X denote its range. Then,

$$\sum_{x \in R_X} p_X(x) = 1$$

5.2 Expectation of a Discrete Random Variable

For a random variable X, its expectation, denoted as $\mathbb{E}[X]$ is defined as,

$$\sum_{x \in R_X} x \cdot p_X(x)$$

Proposition 5.1 (Linearity of Expectation)**.** For any two random variables X and Y, the linearity of expectation tells us that,

$$\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

In more generality, this is stated as follows. Let $X_1, X_2, \cdots, X_n$ be a sequence of random variables. Then,

$$\mathbb{E}\left[\sum_{i=1}^{n}(X_i)\right] = \sum_{i=1}^{n}\mathbb{E}[X_i]$$

5.3 Expectation of a Function of a Random Variable

For a random variable X and any other *real* function $g : \mathbb{R} \to \mathbb{R}$, observe that $g(X)$ is also a random variable. Now, $\mathbb{E}[g(X)]$ can be computed simply as,

$$\mathbb{E}[g(X)] = \sum_{x \in R_X} g(x) \cdot p_X(x)$$

Remark. If $g(x) = x^2$, then the random variable $g(X)$ is more simply denoted as, X^2. Similarly, if $g(x) = ax + b$, then the random variable $g(X)$ is more simply denoted as, $aX + b$ and so on.

Proposition 5.2. For a random variable X, and any $a, b \in \mathbb{R}$,

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$$

5.4 Variance and Standard Deviation of a Discrete Random Variable

For a random variable X, its variance, denoted as $Var(X)$, is defined as,

$$\mathbb{E}\left[(X - \mathbb{E}[X])^2\right]$$

A much more simpler way to compute $Var(X)$ is,

$$Var(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

The standard deviation of a random variable X, denoted as, $SD(X)$, is defined as,

$$SD(X) = \sqrt{Var(X)}$$

Proposition 5.3. For a random variable X, and any $a, b \in \mathbb{R}$,

$$Var(aX + b) = a^2 Var(X)$$

5.5 Cumulative Distribution Function (CDF) of a Random Variable

For a random variable X, its CDF, denoted as F_X, is defined as,

$$F_X(x) = Pr(X \leq x)$$

6 Discrete Distributions

In this section, we will explore some common and popular discrete random variables. Every discrete random variable discussed here is associated with some *random experiment*. We don't really need to explicitly memorise PMF or other aspects of every example discussed here, but rather it suffices to understand the associated random experiment well, using which all the other properties can be derived easily.

6.1 Bernoulli Random Variable

Consider an experiment whose every outcome can be categorized as either a *success* or a *failure*. Such an experiment is called a Bernoulli trial. Let the random variable X be defined as follows,

$$X = \begin{cases} 1, & \text{if outcome is a } \textit{success} \\ 0, & \text{if outome is a } \textit{failure} \end{cases}$$

Then such a random variable X is called a Bernoulli random variable.

6.1.1 PMF of a Bernoulli Random Variable

Let p denote the *success* probability of a Bernoulli trial. Then the PMF of the associated random variable X is defined as,

$$p_X(x) = \begin{cases} p, & \text{if } x = 1 \\ 1 - p, & \text{if } x = 0 \\ 0, & \text{otherwise} \end{cases}$$

We will denote such a Bernoulli random variable as, $X \sim Bernoulli(p)$

6.1.2 Expectation of a Bernoulli Random Variable

For a Bernoulli random variable X,

$$\mathbb{E}[X] = p$$

6.1.3 Variance of a Bernoulli Random Variable

For a Bernoulli random variable X,

$$Var(X) = p \cdot (1 - p)$$

6.2 Binomial Random Variable

Recall the Bernoulli trial discussed above. If such a Bernoulli trial is repeated some n number of times *independently* and we are interested in counting the total number of *successes* in the n trials, then the random variable X, defined as,

$$X = k,$$

if total number of successes in n trials is k, is called a binomial random variable. A binomial random variable X is also denoted as $X \sim Bin(n, p)$, where the *parameters* n and p represent the total number of times a Bernoulli trial has been repeated, and the success probability of each Bernoulli trial, respectively.

6.2.1 PMF of a Binomial Random Variable

For a binomial random variable $X \sim Bin(n, p)$, its PMF is given as follows,

$$p_X(k) = \begin{cases} \binom{n}{below} p^k (1-p)^{(n-k)}, & k \in \{0, 1, 2, \cdots, n\} \\ 0, & \text{otherwise} \end{cases}$$

6.2.2 Expectation of a Binomial Random Variable

Expectation of a binomial random variable $X \sim Bin(n, p)$ is defined as,

$$\mathbb{E}[X] = np$$

6.2.3 Variance of a Binomial Random Variable

Variance of a binomial random variable $X \sim Bin(n, p)$ is defined as,

$$Var(X) = np(1-p)$$

6.2.4 Useful R language commands for Binomial Distribution

Following are simple and useful R language commands related to binomial distribution.

Consider a binomial random variable $X \sim Bin(n, p)$. If we want to generate some y number of realizations of the random variable X, then we can use the following R command.

```
rbinom(y, n, p)
```

Suppose $X \sim Bin(10, 0.5)$, then to generate 500 different realizations, we will use the above command as follows.

```
rbinom(500, 10, 0.5)
```

To compute PMF of a binomial random variable X at k, i.e., to compute $p_X(k) = \Pr(X = k)$, we use following R command.

```
dbinom(k, n, p)
```

Considering again, $X \sim Bin(10, 0.5)$, then to compute $p_X(5) = \Pr(X = 5)$, we will write,

```
dbinom(5, 10, 0.5)
```

If instead of computing PMF, we want to compute CDF of a binomial random variable X at k, i.e., to compute, $F_X(k) = \Pr(X \le k)$, we use the following R command,

```
pbinom(k, n, p)
```

Suppose same as above, $X \sim Bin(10, 0.5)$ and we want to compute $F_X(2) = \Pr(X \le 2)$, then we will write,

```
pbinom(2, 10, 0.5)
```

6.3 Geometric Random Variable

Consider once again some Bernoulli trial. Now suppose we repeat the trial until we observe success for the first time. Then the random variable X, defined as,

$$X = k$$

where k denotes the number of trials including and upto the first success, is called a geometric random variable. Such a geometric random variable X, is also denoted as $X \sim Geom(p)$ where the parameter p denotes the success probability of the corresponding Bernoulli trial.

6.3.1 PMF of a Geometric Random Variable

For a geometric random variable $X \sim Geom(p)$, its PMF is given as follows,

$$p_X(k) = \begin{cases} (1-p)^{k-1}p, & \text{if } k \in \mathbb{Z}_{\geq 0} \\ 0, & \text{otherwise} \end{cases}$$

6.3.2 Expectation of a Geometric Random Variable

Expectation of a geometric random variable $X \sim Geom(p)$ is given as,

$$\mathbb{E}[X] = \frac{1}{p}$$

6.3.3 Variance of a Geometric Random Variable

Variance of a geometric random variable $X \sim Geom(p)$ is given as,

$$Var(X) = \frac{1-p}{p^2}$$

6.3.4 Useful R language commands for Geometric Distribution

Following are some useful R language commands related to a geometric random variable.

For a geometric random variable $X \sim Geom(p)$. Suppose we want to generate some y number of independent realizations of X. Then we can use the following R command.

```
rgeom(y, p)
```

To compute $p_X(k)$, we use the following command,

```
dgeom(k-1, p)
```

Finally, to compute $F_X(k)$, we use the following command,

```
pgeom(k-1, p)
```

6.4 Poisson Random Variable

A Poisson random variable can be thought of as an approximation of a binomial random variable when the parameter n gets very large and the other parameter p gets relatively small. A Poisson distribution is observed in many real life scenarios such as the number of customers walking into a supermarket store which generally follows a Poisson distribution. A Poisson random variable X is also denoted as $X \sim Poisson(\lambda)$, where the parameter $\lambda > 0$ is some constant.

6.4.1 PMF of a Poisson Random Variable

For a Poisson random variable $X \sim Poisson(\lambda)$, its PMF is given as follows,

$$p_X(k) = \begin{cases} \frac{e^{-\lambda}\lambda^k}{k!}, & \text{if } k \in \mathbb{Z}_{\geq 0} \\ 0, & \text{otherwise} \end{cases}$$

6.4.2 Expectation of a Poisson Random Variable

Expectation of a Poisson random variable $X \sim Poisson(\lambda)$ is given as,

$$\mathbb{E}[X] = \lambda$$

6.4.3 Variance of a Poisson Random Variable

Variance of a Poisson random variable $X \sim Poisson(\lambda)$ is given as,

$$Var(X) = \lambda$$

6.4.4 Useful R language commands for Poisson Distribution

Following are some useful R language commands corresponding to a Poisson distribution.

For a Poisson random variable $X \sim Poisson(\lambda)$, if we want to compute $p_X(k)$, then we can use the following R command,

```
dpois(k, λ)
```

To compute $F_X(k)$, we can use the following R command,

```
ppois(k, λ)
```

7 Continuous Random Variables

Definition 7.1 (Continuous Random Variables)**.** A random variable X is called a continuous random variable, if R_X, i.e., the *range* of X, is an uncountably infinite set.

The way we defined probability mass function for a discrete random variable, we can't really define PMF for a continuous random variable the same way. Because there are uncountably infinite many points and if each associated point has a non-zero probability then the total probability would exceed one. So instead what we define for a continuous random variable is called *probability density function (pdf)*.

Remark. For a continuous random variable X, $\Pr(X = x) = 0$ for any $x \in \mathbb{R}$.

7.1 CDF of a Continuous Random Variable

Let X be a continuous random variable, then CDF of X, denoted $F_X(x)$ just as for the discrete case is defined as, for any $x \in \mathbb{R}$,

$$F_X(x) = \Pr(X \leq x)$$

Proposition 7.1. For any continuous random variable X and any $\alpha, \beta \in \mathbb{R}$, we have,

$$\Pr(\alpha \leq X \leq \beta) = F_X(\beta) - F_X(\alpha)$$

7.2 PDF of a Continuous Random Variable

Consider a continous random variable X with CDF $F_X(x)$, then the probability density function (PDF) of X, denoted as, $f_X(x)$ is defined as,

$$f_X(x) = \frac{dF_X(x)}{dx} = F'_X(x)$$

if $F_X(x)$ is differentiable at x.

7.3 Expectation of a Continuous Random Variable

For a continuous random variable X with pdf $f_X(x)$, its expectation is defined very similar to that of a discrete one, just that the *summation* is replaced *integration*,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

7.4 Variance and Standard Deviation of a Continuous Random Variable

Variance of continous random variable X, with pdf $f_X(x)$ is defined as,

$$Var(X) = \int_{-\infty}^{\infty} x^2 f_X(x) dx - (\mathbb{E}[X])^2$$

As before, standard deviation of the random variable X, $SD(X)$, is defined as,

$$SD(X) = \sqrt{Var(X)}$$

7.5 pth Quantile of a Random Variable

Definition 7.2 (pth Quantile of a Random Variable). Consider a random variable X. For any $0 < p < 1$, pth quantile of X, denoted as ϕ_p, is the smallest value such that,

$$F_X(\phi_p) = \Pr(X \leq \phi_p) \geq p.$$

If X is a continuous random variable, then the pth quantile, ϕ_p, is the smallest value such that,

$$F_X(\phi_p) = \Pr(X \leq \phi_p) = p.$$

8 Continuous Distributions

8.1 Uniform Distribution

Perhaps the simplest of the continuous distribution is a uniform distribution. Consider some interval $[a, b]$. The idea of a uniform distribution is that the probability of selecting a number from some sub-interval $[x_1, x_2]$, where, $a \leq x_1 \leq x_2 \leq b$, is directly proportional to its length, $x_2 - x_1$.

8.1.1 PDF of a Uniform Distribution

The idea of a pdf is in my opinion best explained by explaining it for a uniform distribution. Let the random variable X denote the number selected *uniformly* at random from $[a, b]$. Such a random variable is also denoted as $X \sim Unif(a, b)$. Then the probability that the selected number lies in the interval is $[a, b]$ is 1. Now assuming uniform distribution of probability in this whole interval, the probability that the selected number lies in an interval of unit length would be $\frac{1}{b-a}$. Then the probability that the selected number lies in the interval $[x_1, x_2]$, for $a \leq x_1 \leq x_2 \leq b$, would be $\frac{x_2 - x_1}{b-a}$. Such a random variable X is said to have a uniform distribution. The pdf of such a random variable, denoted as, f_X is given as,

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

If you observe, the pdf is in some sense telling us the probability distribution per unit length. This is *not* entirely correct though, but it is perhaps a reasonable intuition to have for now.

8.1.2 Expectation of a Uniformly Distributed Random Variable

For a uniformly distributed random variable X, its expectation is given as,

$$\mathbb{E}[X] = \frac{a+b}{2}$$

8.1.3 Variance of a Uniformly Distributed Random Variable

Variance of a uniformly distributed random variable $X \sim Unif(a, b)$ is given as,

$$Var(X) = \frac{(b-a)^2}{12}$$

8.1.4 Useful R language commands for Uniform Distribution

Following are some useful R language commands related to a uniform distribution.

Suppose we want to generate some y number of a uniformly distributed random variable $X \sim Unif(a, b)$, then we can use the following R command.

```
runif(y, a, b)
```

8.2 Normal Distribution

A normal distribution is perhaps the most used continous distribution. It is defined and characterized by its pdf.

8.2.1 PDF of a Normal Distribution

A random variable X is said to have a *normal* distribution if its PDF, $f_X(x)$, is given as,

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

where, $\sigma > 0$ and $\mu \in \mathbb{R}$.

Such a random variable X is also denoted as $X \sim Norm(\mu, \sigma)$. A random variable $X \sim Norm(0, 1)$ is called a *standard* normal random variable.

8.2.2 Expectation of a Normally Distributed Random Variable

For $X \sim Norm(\mu, \sigma)$, its expectation is given as,

$$\mathbb{E}[X] = \mu$$

8.2.3 Variance of a Normally Distributed Random Variable

Variance of a normally distributed random variable $X \sim Norm(\mu, \sigma)$, is given as,

$$Var(X) = \sigma^2$$

8.2.4 Empirical Results

Let $X \sim Norm(\mu, \sigma)$, then we have following results which holds for any set of measurements that approximately follows normal distribution,

- $\Pr(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68$
- $\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$
- $\Pr(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0.997$

8.2.5 Useful R language commands for Normal Distribution

Following are some useful R language commands corresponding to normal distribution.

Suppose $X \sim Norm(\mu, \sigma)$. Then to generate some y number of independent realizations of X, we can use the following R command.

```
rnorm(y, μ, σ)
```

To compute cumulattive distribution function (cdf), $F_X(k)$, we use the following command,

```
pnorm(k, μ, σ)
```

Using *pnorm* command

Suppose for $X \sim Norm(\mu, \sigma)$, we want to compute $\Pr(a \leq X \leq b)$, then we would need to evaluate,

$$\int_a^b \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{\frac{-(x-\mu)^2}{2\sigma^2}} dx$$

There isn't any closed form or simple trick to evaluate this value thus we prefer using computers to evaluate this for us. **For example,** suppose we want to evaluate $\Pr(-2 \leq X \leq 2)$, where $X \sim Norm(0, 1)$. Then recall that, $\Pr(-2 \leq X \leq 2) = F_X(2) - F_X(-2)$, thus we can use *pnorm* command of R language as follows,

```
print(pnorm(2, 0, 1) - pnorm(-2, 0, 1))
```

which will print the $\Pr(-2 \leq X \leq 2)$, which is 0.9544997.

To compute pth quantile, ϕ_p such that $F_X(\phi_p) = p$, of a normally distributed random variable, we can use the following R command,

```
qnorm(p, μ, σ)
```

8.3 Gamma Distribution

A random variable X is said to have a *gamma* distribution if its PDF, $f_X(x)$, is given as,

$$f_X(x) = \begin{cases} \frac{x^{\alpha-1}e^{-x/\beta}}{\beta^\alpha \Gamma(\alpha)}, & \text{if } 0 \leq x < \infty \\ 0, & \text{otherwise} \end{cases}$$

where $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1}e^{-x}$ and α and β are associated non-negative parameters.

The function $\Gamma(\alpha)$ is called gamma function which among other things can also be thought of as an extension of the factorial function defined for the non-negative integers to the set of all non-negative real numbers. Two facts about the gamma function are, $\Gamma(1) = 1$ and $\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$. The second fact implies that if n is a non-negative integer then $\Gamma(n) = (n - 1)!$.

8.3.1 Expectation of a Gamma Distributed Random Variable

If X is a gamme distributed random variable with associated parameters $\alpha > 0$ and $\beta > 0$, then,

$$\mathbb{E}[X] = \alpha\beta$$

8.3.2 Variance of a Gamma Distributed Random Variable

If X is a gamme distributed random variable with associated parameters $\alpha > 0$ and $\beta > 0$, then,

$$Var(X) = \alpha\beta^2$$

8.3.3 Useful R language commands for Gamma Distribution

Following are some useful R language commands corresponding to a Gamma distribution. For a gamma-distributed random variable X with parameters $\alpha > 0$ and $\beta > 0$, then to compute $F_X(k)$, we can use,

```
pgamma(k, α, 1/β)
```

and to compute $p_{th} quantile$ we can use,

```
qgamma(p, α, 1/β)
```

8.4 Chi-Square Distribution

Let ν be a positive integer. Then a random variable X is said to follow *chi-square* (χ^2) distribution with ν *degrees of freedom* if and only if X is a gamma-distributed random variable with parameters $\alpha = \nu/2$ and $\beta = 2$.

8.4.1 Expectation of Chi-Squared Distributed Random Variable

Let X be a chi-squared distributed random variable with ν degrees of freedom, then,

$$\mathbb{E}[X] = \nu$$

8.4.2 Variance of a Chi-Squared Distributed Random Variable

Let X be a chi-squared distributed random variable with ν degrees of freedom, then,

$$Var(X) = 2\nu$$

8.4.3 Useful R language commands for Chi-Squared Distributed Random Variable

If X is a χ^2 distributed random variable with ν degrees of freedom then to compute $F_X(k)$, we can use,

```
pchisq(k, ν)
```

To compute p_{th} quantile, we can use,

```
qchisq(p, ν)
```

8.5 Exponential Distribution

A random variable X is said to have an *exponential* distribution if and only if X follows gamma distribution with parameter $\alpha = 1$, i.e., PDF of X is given as,

$$f_X(x) = \begin{cases} \frac{1}{\beta} e^{-x/\beta}, & 0 \leq x < \infty \\ 0, & \text{elsewhere} \end{cases}$$

8.5.1 Expectation of an Exponentially Distributed Random Variable

If X is an *exponentially* distributed random variable with associated parameter β, then,

$$\mathbb{E}[X] = \beta$$

8.5.2 Variance of an Exponentially Distributed Random Variable

If X is an *exponentially* distributed random variable, then,

$$Var(Y) = \beta^2$$

8.5.3 Useful R language commands for Exponentially Distributed Random Variable

If X is an exponentially distributed random variable with parameter β then to compute $F_X(k)$, we can use,

```
pexp(k, 1/β)
```

To compute p_{th} quantile, we can use,

```
qexp(p, 1/β)
```

9 Some Sampling Distributions Related to the Normal Distribution

9.1 Some Useful Theorems

Theorem 9.1. *Suppose $Y_1, Y_2, \cdots, Y_n$ are iid normally distributed random variables with mean μ and standard deviation σ. Then,*

$$\overline{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$$

follows normal distribution with mean μ and standard deviation $\sigma/\sqrt{n}$, i.e., $\overline{Y} = Norm(\mu, \sigma/\sqrt{n})$.

Theorem 9.2. *Suppose $Y_1, Y_2, \cdots, Y_n$ are iid normally distributed random variables with mean μ and standard deviation σ. Then, for $i = 1, \cdots, n$, considering $Z_i = (Y_i - \mu)/\sigma$,*

$$\sum_{i=1}^{n} Z_i^2 = \sum_{i=1}^{n} \left(\frac{Y_i - \mu}{\sigma}\right)^2$$

follows χ^2 distribution with n df.

Theorem 9.3. *Suppose $Y_1, Y_2, \cdots, Y_n$ are iid normally distributed random variables with mean μ and standard deviation σ. Then,*

$$\frac{(n-1)S^2}{\sigma^2} = \frac{1}{\sigma^2}\sum_{i=1}^{n} \left(Y_i - \overline{Y}\right)^2$$

follows χ^2 distribution with $n-1$ df. In fact, S^2 and $\overline{Y}$ are independent random variables.

9.2 t Distribution

If Z is a standard normal random variable and W is a χ^2 distributed random variable with ν *df*, then the random variable T defined as follows,

$$T = \frac{Z}{\sqrt{W/\nu}}$$

follows t distribution with ν *df.*

9.2.1 Useful R language commands for t Distributed Random Variable

If Y is a t-distributed random variable with ν *df*, then to compute $F_Y(k) = \Pr(Y \le k)$, we can use,

```
pt(k, ν)
```

And to compute the p_{th} quantile, we can use,

```
qt(p, ν)
```

9.3 F Distribution

Suppose W_1 is a χ^2 distributed random variable with ν_1 *df* and W_2 is a χ^2 distributed random variable with ν_2 *df*. Then the random variable F defined as follows,

$$F = \frac{W_1/\nu_1}{W_2/\nu_2}$$

follows F distribution with ν_1 numerator *df* and ν_2 denominator *df*.

Assume we have two independent populations with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2, respectively. Now suppose we have two (independent) random samples of sizes n_1 and n_2, respectively and let S_1^2 and S_2^2 be sample variances corresponding to population 1 and 2 respectively. Then,

$$F = \frac{\left(\frac{(n_1-1)S_1^2}{\sigma_1^2}\right)/(n_1-1)}{\left(\frac{(n_2-1)S_2^2}{\sigma_2^2}\right)/(n_2-1)} = \left(\frac{S_1^2}{S_2^2}\right)\left(\frac{\sigma_2^2}{\sigma_1^2}\right)$$

follows F distribution with $n_1 - 1$ numerator *df* and $n_2 - 1$ denominator *df*.

9.3.1 Useful R language commands for F Distributed Random Variable

If Y is an F-distributed random variable with ν_1 numerator *df* and ν_2 denominator *df*, then to compute $F_Y(k)$, we can use,

```
pf(k, ν1, ν2)
```

and to compute the p_{th} quantile, we can use,

```
qf(p, ν1, ν2)
```

10 Central Limit Theorem

A very useful theorem in the probability theory is this so-called Central Limit Theorem.

Theorem 10.1. *Let $X_1, X_2, \cdots, X_n$ be independent and identically distributed (iid) random variables, where $\mathbb{E}[X_i] = \mu$ and $Var(X_i) = \sigma^2$, for $i \in \{1, 2, \cdots, n\}$. Then, the random variable $\overline{X}$, defined as,*

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

has approximately *a normal distribution with mean μ and variance σ^2/n,*
i.e., $\overline{X} \sim Norm(\mu, \sigma/\sqrt{n})$ as $n \to \infty$.

Considering $Z = \frac{\overline{X}-\mu}{\sigma/\sqrt{n}}$, the theorem can be restated as, The random variable Z has approximately *a* standard *normal distribution as $n \to \infty$, i.e., $Z \sim Norm(0, 1)$.*

10.1 Approximation of a Binomial Distribution as a Normal Distribution

Recall that a binomial random variable $X \sim Bin(n, p)$ counts the number of successes in a sequence of n independent and identically distributed Bernoulli trials, i.e., Considering $X_i =$

$Bernoulli(p)$, then we can write,

$$X = \sum_{i=1}^{n} X_i$$

Then by the Central Limit Theorem above, we have that, X/n has approximately normal distribution.

The key point to remember here is that this approximation becomes finer and finer for larger values of n. In fact, if we have, $0 < p \pm 3\sqrt{pqn} < 1$ or equivalently, if we have,

$$n > 9 \cdot \frac{max(p, q)}{min(p, q)}$$

where, $q = 1 - p$, then the binomial random variable will have a better approximation to a normal distribution.

To illustrate this, Consider $X \sim Bin(n, 0.001)$. Then by the above discussion, we would need $n > 8991$ to have a better approximation to normal distribution. Now consider the following R code that plots number of successes as a histogram for the above binomial random variable X.

```
successes <- c()
for (i in 1:10000){
  outcomes <- sample(c(0, 1), 2000, rep = TRUE, prob = c(0.999, 0.001))
  successes <- append(successes, sum(outcomes))
}
hist(main = "Central Limit Theorem", successes)
```

Run the above code and see the output. Now try to increase the value 2000 and see how the graph changes. You will observe that the graph starts appearing like a bell curve once you reach around 9000 or higher.